

# Route decision: Port of Shanghai → Port of Los Angeles

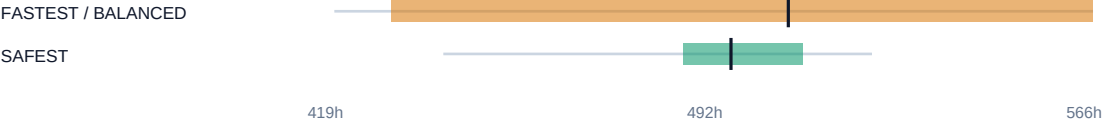
PORT-SHANGHAI → PORT-LOSANGELES · mode sea (STRICT) · cargo general · priority normal · scenarios: LA\_PORT\_STRIKE · planned 2026-09-29 13:47 UTC

**Recommendation:** FASTEST / BALANCED, SEA via direct. Expected arrival in **430 h** (17.9 days); plan for 507 h (p85), worst case 566 h (p95). Cost: freight CN¥11,597. Peak leg risk 90%.

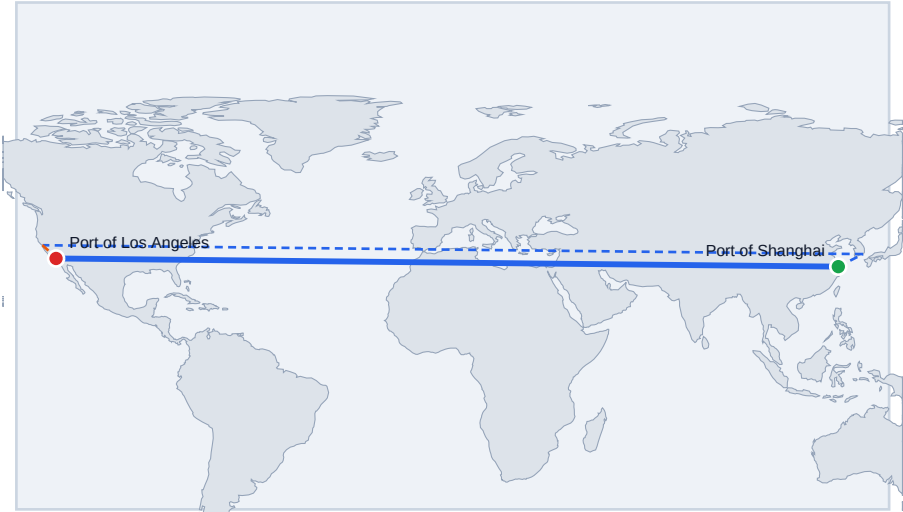
## Options

| Option             | Mode | Via    | Plan h | p50 h | p85 h | p95 h | Freight   | Risk |
|--------------------|------|--------|--------|-------|-------|-------|-----------|------|
| FASTEST / BALANCED | SEA  | direct | 419    | 430   | 507   | 566   | CN¥11,597 | 90%  |
| SAFEST             | SEA  | direct | 440    | 486   | 496   | 509   | CN¥18,270 | 90%  |

ETA bands (bar p50–p95, tick p85, line plan → fully-correlated worst case)



## Routes



Recommended route solid, alternatives dashed. Blue sea, amber air, green rail, orange road.

FASTEST / BALANCED: SEA via direct

SEA via no chokepoints; expected 429.8 h (p85 506.7 h, p95 565.7 h), CN¥11,597 freight, 0 handoffs. 56.5 h faster than the next option. CN¥6,675 cheaper (freight) than the lowest-cost alternative. Includes 120.0 h announced scenario delay.

| # | Mode | To                  | km     | Hours | Delay p85 | Cost      | Risk | Signal                                                                               |
|---|------|---------------------|--------|-------|-----------|-----------|------|--------------------------------------------------------------------------------------|
| 1 | SEA  | Port of Los Angeles | 10,457 | 418.8 | +88.0     | CN¥10,543 | 90%  | SCENARIO: Terminal labor strike. Picket lines at all major berths. Throughput at 0%. |

Transit 298.8 h + transfers 0.0 h + announced scenario delay 120.0 h = plan 418.8 h. Model delay buffer p50/p85/p95: +11.1 / +88.0 / +146.9 h. Route band p50 429.8 h, p85 506.7 h, p95 565.7 h (independent legs); 565.7 h if all legs are late together. Costs: linehaul CN¥10,543, transfers CN¥0.00, disruption surcharge CN¥1,054.

Why the slowest leg (PORT-SHANGHAI → PORT-LOSANGELES, p85 208.0 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



SAFEST: SEA via direct

SEA via no chokepoints; expected 486.3 h (p85 495.6 h, p95 509.4 h), CN¥18,270 freight, 2 handoffs. 56.5 h slower than FASTEST. CN¥6,675 more (freight) than FASTEST. Includes 120.0 h announced scenario delay.

| # | Mode    | To                           | km    | Hours | Delay p85 | Cost      | Risk | Signal                                                                               |
|---|---------|------------------------------|-------|-------|-----------|-----------|------|--------------------------------------------------------------------------------------|
| 1 | SEA     | Port of Busan                | 824   | 23.5  | +26.6     | CN¥830.92 | 0%   |                                                                                      |
| 2 | SEA     | Port of Oakland              | 9,080 | 259.4 | +27.5     | CN¥9,154  | 0%   |                                                                                      |
| 3 | handoff | Port of Oakland              |       | 14.0  | +0.0      | CN¥1,680  | 8%   |                                                                                      |
| 4 | ROAD    | San Francisco Logistics Park | 13    | 0.2   | +3.6      | CN¥101.29 | 0%   |                                                                                      |
| 5 | ROAD    | LA Logistics Cluster         | 559   | 8.2   | +3.6      | CN¥4,510  | 0%   |                                                                                      |
| 6 | ROAD    | Port of Los Angeles          | 35    | 120.5 | +0.0      | CN¥283.85 | 90%  | SCENARIO: Terminal labor strike. Picket lines at all major berths. Throughput at 0%. |
| 7 | handoff | Port of Los Angeles          |       | 14.0  | +0.0      | CN¥1,680  | 8%   |                                                                                      |

Transit 291.9 h + transfers 28.0 h + announced scenario delay 120.0 h = plan 439.9 h. Model delay buffer p50/p85/p95: +46.4 / +61.4 / +83.0 h. Route band p50 486.3 h, p85 495.6 h, p95 509.4 h (independent legs); 522.9 h if all legs are late together. Costs: linehaul CN¥14,880, transfers CN¥3,361, disruption surcharge CN¥28.37.

Why the slowest leg (PORT-BUSAN → PORT-OAKLAND, p85 27.5 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



Method: legs are priced with gradient-boosted quantile models (p50/p85/p95) trained on synthetic delays fitted to published UNCTAD, World Bank and STB statistics; FASTEST plans on p50, BALANCED on p85 (plus cargo carrying cost when a value is given), SAFEST on p95 plus 240 risk-hours per unit of threat. Route bands assume independent legs. Sea distances are great-circle between chokepoint waypoints. Currency conversion: live (open.er-api.com) as of Tue, 29 Sep 2026 00:02:31 +0000.